

MUS 442 · SENIOR PROJECT II · FINAL PRESENTATION

The Shift

A VR sound narrative about maladaptive daydreaming, alienation, and the loops we build to escape them.

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THE SHIFT

**THE COLORFUL
PLAYGROUND**

The Question

How can VR and interactive sound make the relationship between alienation, escape, repetition, and creative agency physically perceptible?

MUS 441 — the starting thesis

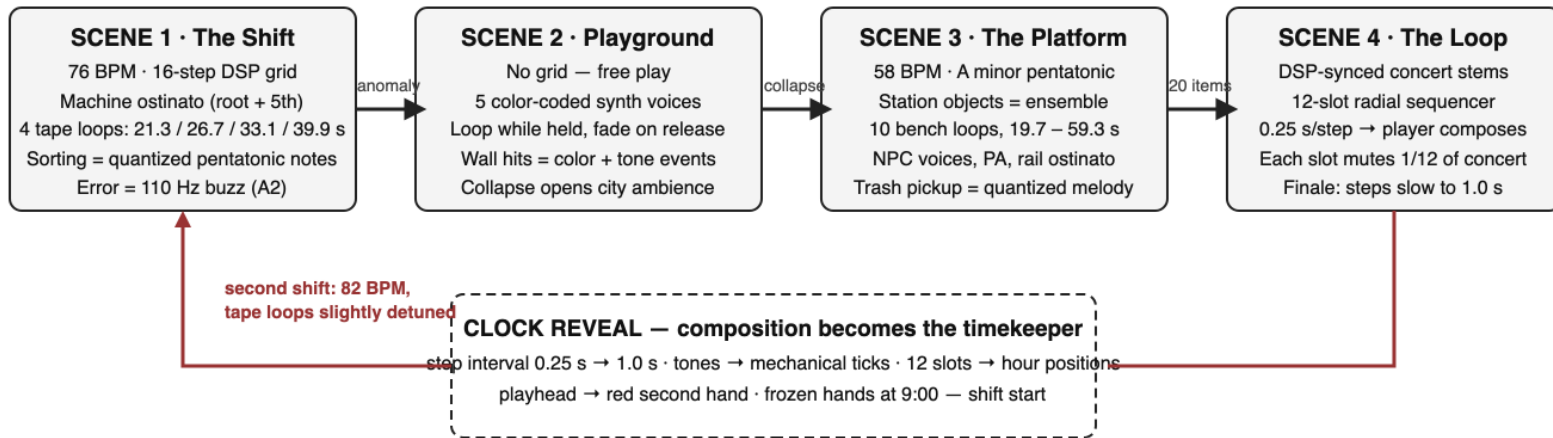
A binary contrast: the gray factory versus the colorful daydream. Escape as liberation.

MUS 442 — the final thesis

A circle: the daydream develops its own routines, and the player's most creative act — composing a loop — rebuilds the factory clock. Imagination is shaped by the reality it escapes.

One Loop, Four Scenes

The Shift — Musical Arc of the Four-Scene Loop



All scenes share one pitch language (minor pentatonic rooted on A) and sample-accurate AudioSettings.dspTime scheduling.

Sorting becomes throwing, throwing becomes cleaning, cleaning becomes composing — and the composition becomes a clock.

Scene 1 • The Shift — labor as score

- Sort 30 red/blue cubes from a conveyor into matching bins — a quota, a wall clock, a score display.
- Correct sorts play quantized pentatonic notes; errors buzz at 110 Hz. Even the reward belongs to the factory.
- The score is generative: machine recordings sliced at runtime into an ostinato and four incommensurate tape loops (Eno).
- A green anomaly cube — unclassifiable — opens the daydream.

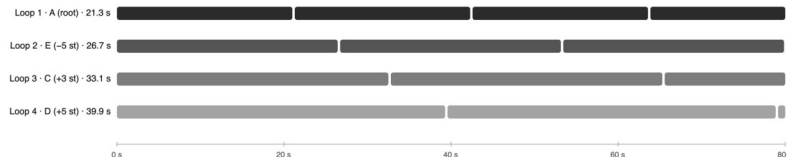
Scene 1 — Factory Score: Ostinato Grid and Incommensurate Tape Loops

Machine ostinato — one bar of 16 sixteenth-steps at 76 BPM (step = 0.197 s)



Percussive transients sliced at runtime from factory machine recordings; root and fifth alternate, octave accent closes the bar.

Four tape loops — periods never align (shown over 80 seconds)



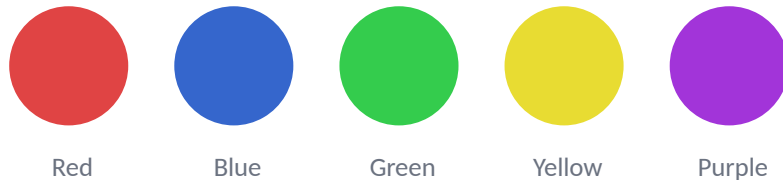
Each loop is anchored to a different machine object in the room. Like Music for Airports, the four periods are incommensurate: the texture never repeats exactly.

76 BPM

sample-accurate DSP grid • second shift runs at 82,
slightly detuned

Scene 2 • The Playground — color as timbre

- Twenty cubes, five colors — each color is a synth voice that sounds while held and fades on release.
- Throwing a cube repaints the wall and plays its tone: the room is a temporary audiovisual composition.
- No grid, no tempo, no quota — the only scene with full musical freedom.
- But play has consequences: twenty wall hits collapse the room into procedural city ambience.



5 colors × 4 cubes = 20 instruments

grab → loop • release → fade • impact → repaint + tone

Five source loops loudness-normalized so no color dominates; impact velocity scales the response.

Scene 3 • The Platform — a metro ensemble

Scene 3 — The Station as Ensemble (58 BPM, A minor pentatonic)

Layer	Sound source	Time structure	Pitch content
1 • Rail ostinato	Track objects	16-step grid, sparse hits	Deep metallic A, low register
2 • Bench pads	10 Metalbench objects	Periods 19.7 → 59.3 s, free-running	Pentatonic dyads (table below)
3 • Lamp drone	Fluorescent lighting fixtures	Continuous hum	A2 + E3 + A3 (root, 5th, octave)
4 • Trash-can ticks	Each MetalTrashCan	Sparse 2-step pattern from index	Pentatonic, low register
5 • NPC voices	Each commuter body	Periods 11.3 / 13.7 / 17.9 / 19.3 / 23.7 / 29.3 s	Hum, murmur, whisper
6 • Tunnel rumble	Tunnel mouths	Continuous, distance-attenuated	Unpitched low noise
7 • PA announcements	Platform speakers	Triggered by train events + idle timer	TTS speech, ducked mix
8 • Trash melody	Player action (cart deposit)	Quantized to next 16th step	Rising pentatonic line, 2 octaves

Bench dyads (semitones above A3 → pitches):

{0, 7} = A3+E4

{3, 12} = C4+A4

{5, 12} = D4+A4

{7, 15} = E4+C5

{10, 19} = G4+E5

Every layer is scheduled on the same dspTime grid but repeats at its own incommensurate period — coherent, never identical, like the commuter cycle itself.

- 58 BPM, A minor pentatonic — benches, lamps, trash cans, rails, and commuters all carry musical layers.
- Routine returns inside the fantasy: trains cycle, NPCs commute, the player voluntarily cleans.
- Each collected trash item plays the next quantized melody note; twenty items complete the scene.

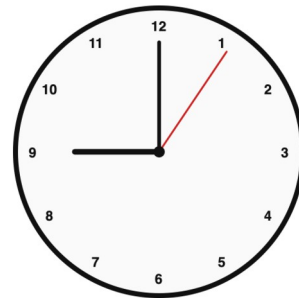
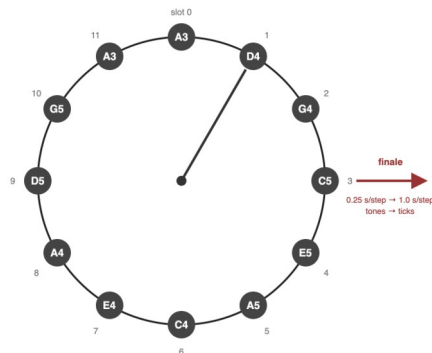
Scene 4 • The Concert / The Loop

- An outdoor concert: DSP-synchronized stems split between stereo playback and 3D stage objects.
- A performing band and a fourteen-person audience react to what the player does.
- The final instrument: a twelve-slot radial sequencer. Placing each sound orb adds a voice to the player's loop — and mutes one twelfth of the concert.
- Complete the the ring, and the spectacle has been fully replaced by the player's own composition.

Scene 4 — The Radial Sequencer is a Clock Face in Disguise

Sequencer mode — step 0.25 s, six synth timbres

Clock mode — one tick per second, hands frozen at 9:00



12 slots • 0.25 s per step • 6 procedural timbres

All tones synthesized at load — A minor pentatonic above A3

The Clock Reveal

The loop the player built begins to slow... until one step lasts exactly one second.

0.25 s → 1.0 s

the playful loop decelerates
to clock time

tones → ticks

the player's voices are
replaced by tick / tock

slots → hours

colors drain; numbers 1-12
fade in; the playhead
becomes the red second
hand

9:00

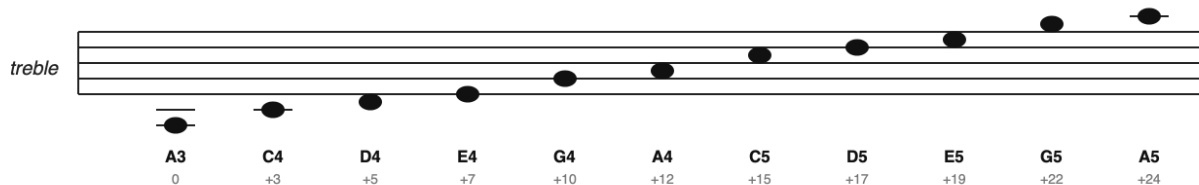
frozen hands appear — shift
start. Scene 1 reloads, faster
and detuned.

The ending is not a message delivered to the player — it is an action the player performs.

One Musical Language, Four Relationships

Shared Pitch Reservoir — A Minor Pentatonic, A3 to A5

Semitones above A3 (220 Hz): 0 · 3 · 5 · 7 · 10 · 12 · 15 · 17 · 19 · 22 · 24 — used by the factory melody, metro ensemble, and sequencer orbs



Scene 1 tape loops transpose the same set below the root: A (0), E (-5), C (+3), D (+5). The error buzz is A2 (110 Hz); the lamp drone stacks A2 + E3 + A3.

- Scene 1 the factory composes — the player obeys its grid
- Scene 2 the player improvises — no grid at all
- Scene 3 the world performs — the player joins the ensemble
- Scene 4 the player composes — and the system captures the result

Under the Hood

4

build scenes

83

C# scripts

~ 16k

lines of C#

100%

generative score systems

Platform

- Unity 6 · Meta Quest 3S · OpenXR · XR Interaction Toolkit
· hand tracking + controllers
- Scenes assembled by procedural systems and custom editor tools, not hand placement
- Hybrid NPC navigation: NavMesh for wandering, authored waypoint chains through train doors

Quest-only problems, solved

- Asset pack shipped no colliders → runtime physics bootstrap
- NPCs floated without shadows → procedural blob shadows + light-probe tooling
- Unstable grabs, LOD pop-outs, rig tilt → runtime repair + on-device diagnostics

What MUS 442 Added

Narrative

- Two scenes became a four-scene circle
- Metro station replaced the earlier Scene 3 direction
- Concert finale designed around the sequencer and clock reveal

Systems

- Cyclic train + NPC boarding system
- Trash-cleanup gameplay with following cart
- Responsive band and fourteen-person audience

Music

- Generative factory and metro scores (Eno tape-loop logic)
- Street ambience and Scene 2 audio polish
- Cross-scene loudness master pass on a common RMS reference



The factory returns.

It sounds different — because the player has heard what lies inside its rhythm.

Demo: the full loop on Meta Quest 3S

factory → playground → platform → concert → clock → second shift

Thank you. — Semiha Paksoy